

☆ Electrical Representation of Encoder output:

⇒ The encoder is a A to D Converter which converts the quantized samples into binary data. In order to represent the binary data electrically the following methods are used.

① ON-OFF Signalling $\begin{cases} 1 \rightarrow +V \\ 0 \rightarrow 0V \end{cases}$

② NRZ Signalling $\begin{cases} 1 \rightarrow +V \\ 0 \rightarrow -V \end{cases}$
(or)
Bipolar Signalling

③ RZ Signalling $\begin{cases} 1 \rightarrow \begin{cases} \frac{T_b}{2} \rightarrow +V \\ \frac{T_b}{2} \rightarrow 0V \end{cases} \\ 0 \rightarrow 0V \end{cases}$

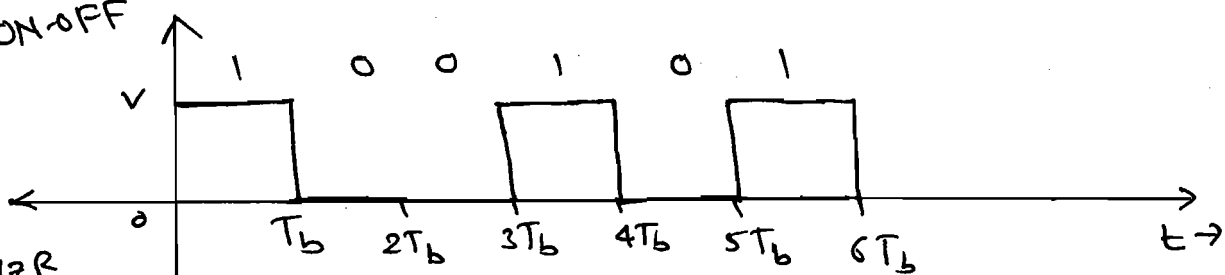
④ Differential Encoding $\begin{cases} 1 \rightarrow \text{Prev. level} \\ 0 \rightarrow \overline{\text{prev. level}} \end{cases}$

⇒ e.g. 100101

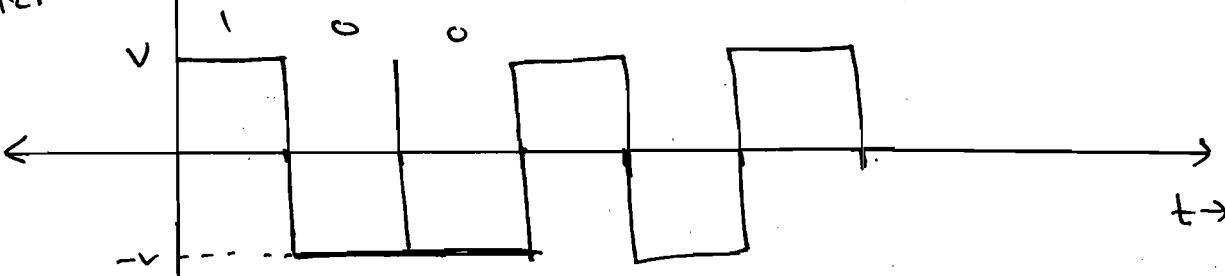
① $\xrightarrow{\underline{0}}$ 010011 → Differential encoded data
Ref. bit

② $\xrightarrow{\underline{1}}$ 101100 → Differential encoded data.

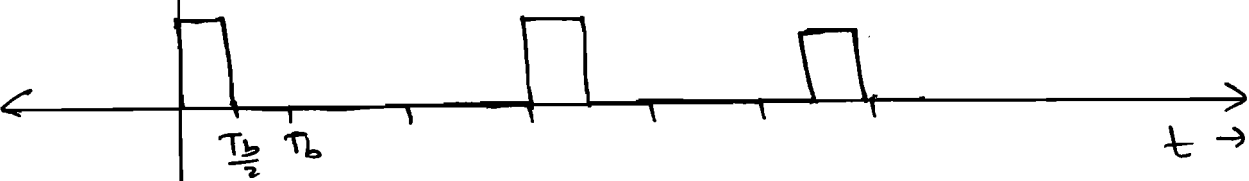
① ON-OFF



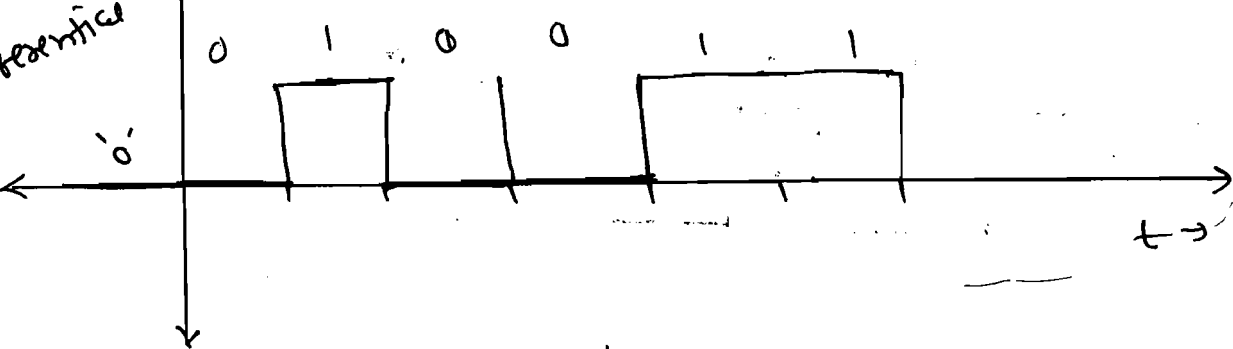
② NRZ



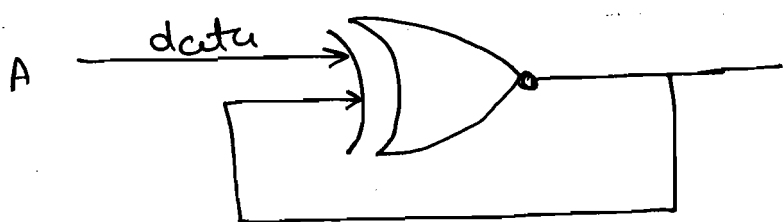
③ RZ



④ Differential



⇒ Differential encoding:



⇒ Most widely used method is NRZ which is used in PCM, DPCM, DM, PSK & FSK.

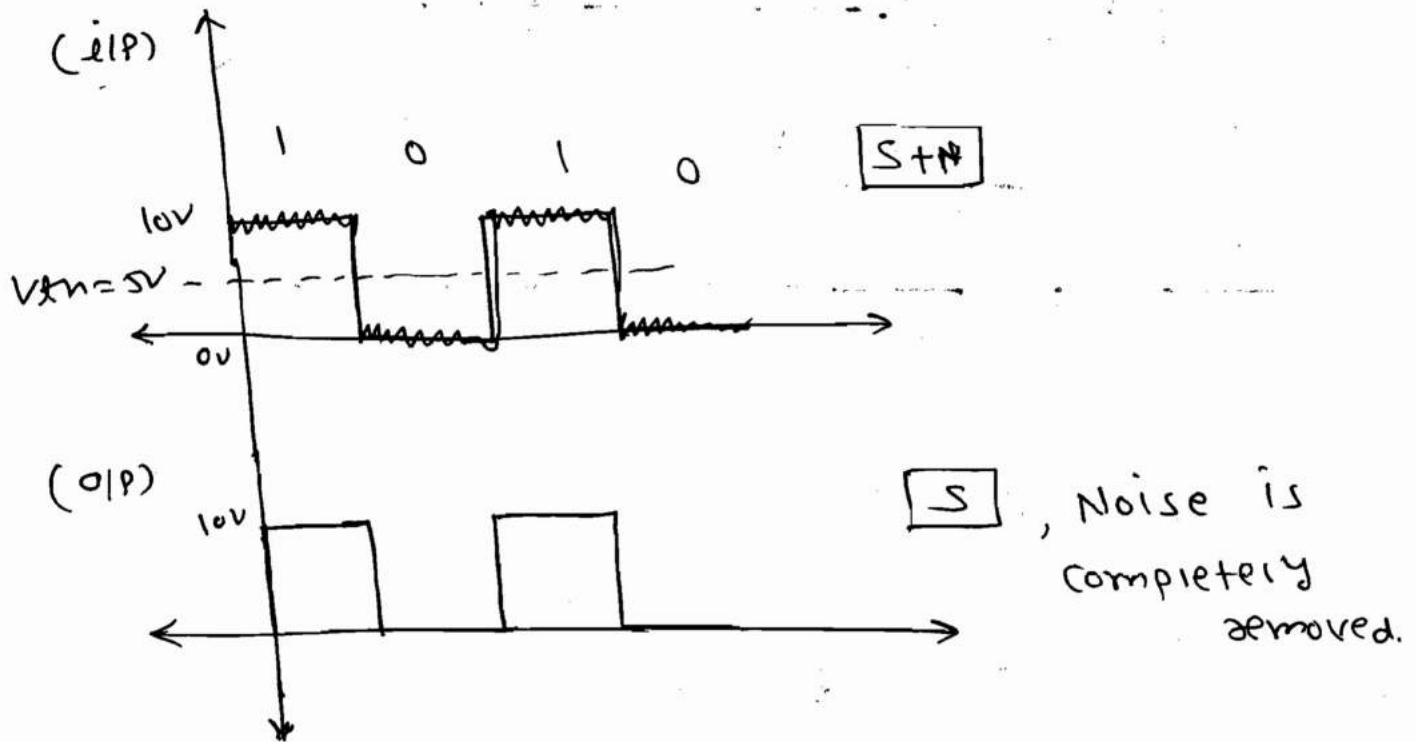
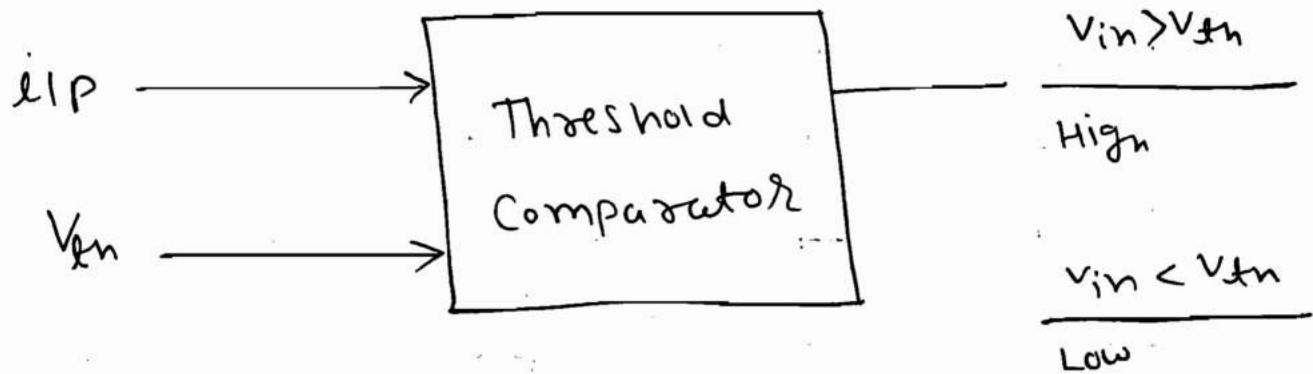
⇒ ON-OFF signaling is used in ASK.

⇒ Differential encoding method is used in DPSK.

is used in FSK

* Regenerative Repeater:

⇒



⇒ When effect of noise is very very large it is possible to get an error.

⇒ The input to the regenerative repeater is binary data affected by the noise. If the noise level is very low signal is regenerated. If the noise level is very

Possible to correct the error using Parity bits. So, the effect of noise is negligible in digital Communication.

⇒ At the receiver threshold Comparator is used as the regenerative circuit. to eliminate the noise at the i/p of receiver.

⇒ Decoder is a digital to analog Converter which converts binary data into samples.

⇒ The LPF is used to reconstruct the signal from sample. But the i/p to the LPF is the sampled signal with quantization error. Due to this error signal distortion occurs and this distortion is called as the Quantization Noise.

⇒ To reduced the quantization error the stepsize should be decrease (or) the no. of levels are increase. This is possible only by increasing n . and As n increases the bit rate ($R_b = n \times \text{samp. rate}$), $BW = R_b$ and Probability of error (P_e) also increases.